One way to achieve simple uncertainty estimates from standard networks is to use the stochastic regularization techniques

What is regularization?



In Bayesian neural networks:

we can say a regularization is implicitly done by “Model Averaging”!

throughout training (and inference)



simple approach based on the following observation:

Modern deep networks are not fully-deterministic actually!



Simple&Effective Techniques: Stochastic Regularization

Stochastic Regularization as Variational Inference

- Stochastic regularization techniques effectively puts a distribution on the weights of neural network, for instance in a multiplicative manner
- Original deterministic parameters: $\boldsymbol{\theta}_t$
- Let us define $\boldsymbol{\theta}'_t = Z_t \boldsymbol{\theta}_t$ and assume $Z_t \sim P(Z)$
- Thus, $\boldsymbol{\theta}'_t$ is not deterministic anymore (use change of variables): $\boldsymbol{\theta}'_t \sim P(\boldsymbol{\theta}/Z)/\boldsymbol{\theta}$

$$L_{VI} = \max_{\boldsymbol{\omega}} \sum \left(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \sum_{i=1}^{n} \log P(y_i|\mathbf{x}_i, \boldsymbol{\theta}) \right) - D_{KL}(Q_{\boldsymbol{\omega}}(\boldsymbol{\theta}) \parallel P(\boldsymbol{\theta}))$$

$$L_{softmax} = \sum -logP(y_i|x_i, \boldsymbol{\theta}) \quad + \quad \|\boldsymbol{\theta}\|_2^2$$

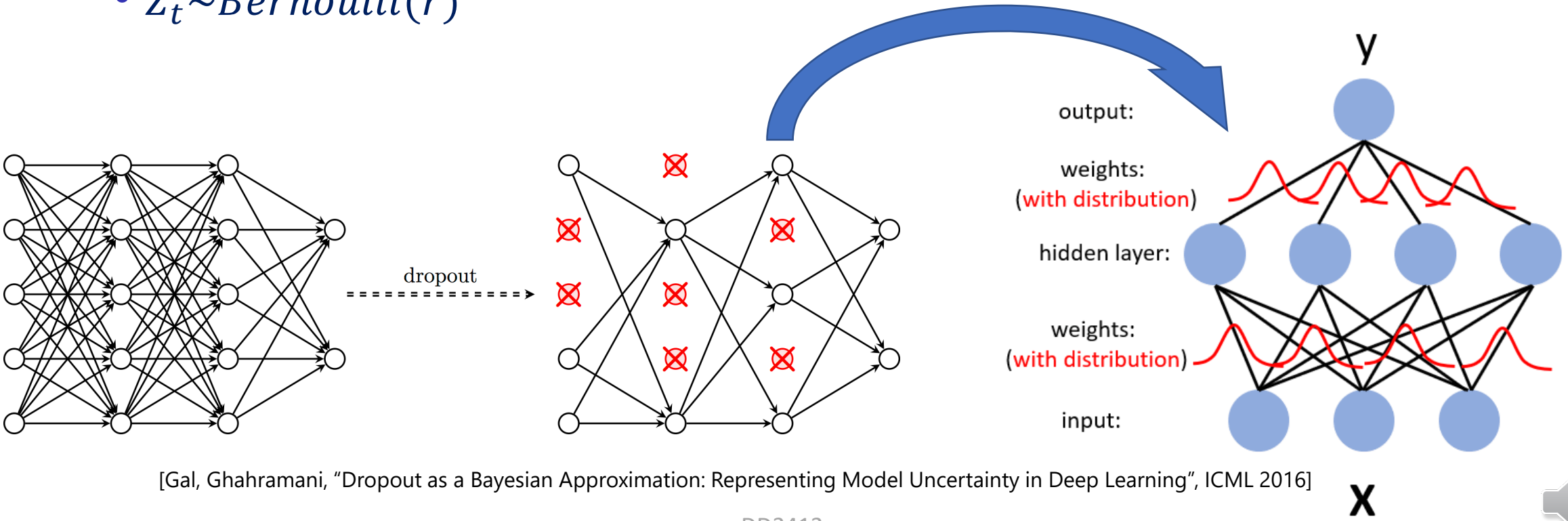
[Gal, Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning", ICML 2016]

Simple&Effective Techniques: Stochastic Regularization

DropOut as Variational Inference: idea

- Stochastic regularization techniques effectively puts a distribution on the weights of neural network, for instance in a multiplicative manner
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- $Z_t \sim Bernoulli(r)$



[Gal, Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning", ICML 2016]

Simple&Effective Techniques: Stochastic Regularization

DropOut as Variational Inference: idea

- Assumptions
  - factorized prior
  - monte carlo expectation
  - …

→Equivalent to variational approximation with a standard Gaussian prior on individual parameters

[Gal, Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning", ICML 2016]

What does this mean?

Simple&Effective Techniques: Stochastic Regularization

DropOut as Variational Inference: inference

$$P(y|x,D) = \int P(y|x,D,\theta)P(\theta|D,x)d\theta = \int P(y|x,\theta)P(\theta|D)d\theta$$

- Simple procedure for obtaining uncertainty from dropout networks:
  1. Sample from dropout network with different masks
     - $\{Z_1, \dots, Z_S\} \sim P(Z)$
     - $\{\boldsymbol{\theta}'_1, \dots, \boldsymbol{\theta}'_S\} = \{Z_i\boldsymbol{\theta}\}_{i=1:S}$
  2. The average is the mean prediction
     - $\hat{y} = E_i[f\ (\boldsymbol{x}; \boldsymbol{\theta}'_i)]$
  3. The variance is the uncertainty
     - $\hat{\sigma}^2 = E_i[(f(\boldsymbol{x}; \boldsymbol{\theta}'_i) - \hat{y})^2]$

[Gal, Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning", ICML 2016]

Simple&Effective Techniques: Stochastic Regularization

DropOut as Variational Inference: Results

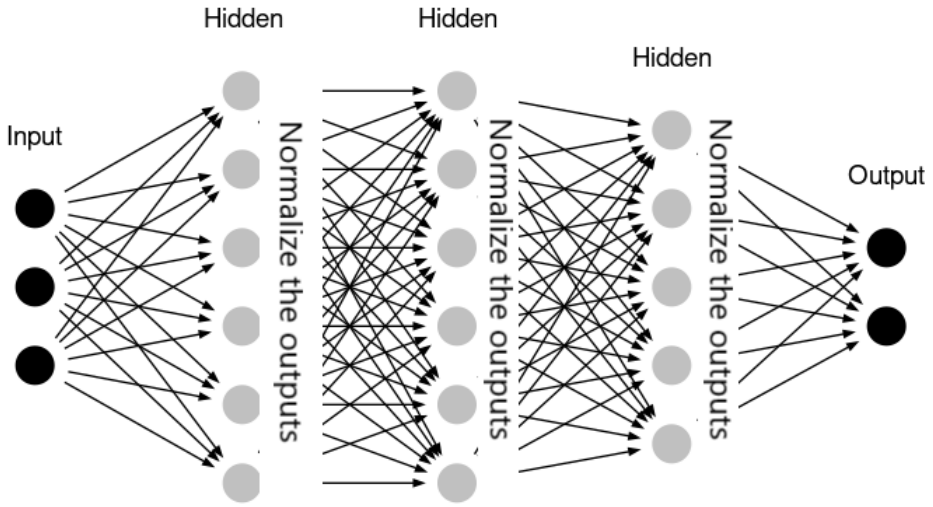
			Avg. Test RMSE and Std. Errors			Avg. Test LL and Std. Errors		
Dataset	N	Q	**VI**	**PBP**	**Dropout**	**VI**	**PBP**	**Dropout**
Boston Housing	506	13	4.32 ±0.29	3.01 ±0.18	**2.97 ±0.19**	-2.90 ±0.07	-2.57 ±0.09	**-2.46 ±0.06**
Concrete Strength	1,030	8	7.19 ±0.12	5.67 ±0.09	**5.23 ±0.12**	-3.39 ±0.02	-3.16 ±0.02	**-3.04 ±0.02**
Energy Efficiency	768	8	2.65 ±0.08	1.80 ±0.05	**1.66 ±0.04**	-2.39 ±0.03	-2.04 ±0.02	**-1.99 ±0.02**
Kin8nm	8,192	8	**0.10 ±0.00**	**0.10 ±0.00**	**0.10 ±0.00**	0.90 ±0.01	0.90 ±0.01	**0.95 ±0.01**
Naval Propulsion	11,934	16	**0.01 ±0.00**	**0.01 ±0.00**	**0.01 ±0.00**	3.73 ±0.12	3.73 ±0.01	**3.80 ±0.01**
Power Plant	9,568	4	4.33 ±0.04	4.12 ±0.03	**4.02 ±0.04**	-2.89 ±0.01	-2.84 ±0.01	**-2.80 ±0.01**
Protein Structure	45,730	9	4.84 ±0.03	4.73 ±0.01	**4.36 ±0.01**	-2.99 ±0.01	-2.97 ±0.00	**-2.89 ±0.00**
Wine Quality Red	1,599	11	0.65 ±0.01	0.64 ±0.01	**0.62 ±0.01**	-0.98 ±0.01	-0.97 ±0.01	**-0.93 ±0.01**
Yacht Hydrodynamics	308	6	6.89 ±0.67	**1.02 ±0.05**	1.11 ±0.09	-3.43 ±0.16	-1.63 ±0.02	**-1.55 ±0.03**
Year Prediction MSD	515,345	90	9.034 ±NA	8.879 ±NA	**8.849 ±NA**	-3.622 ±NA	-3.603 ±NA	**-3.588 ±NA**

[Gal, Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning", ICML 2016]

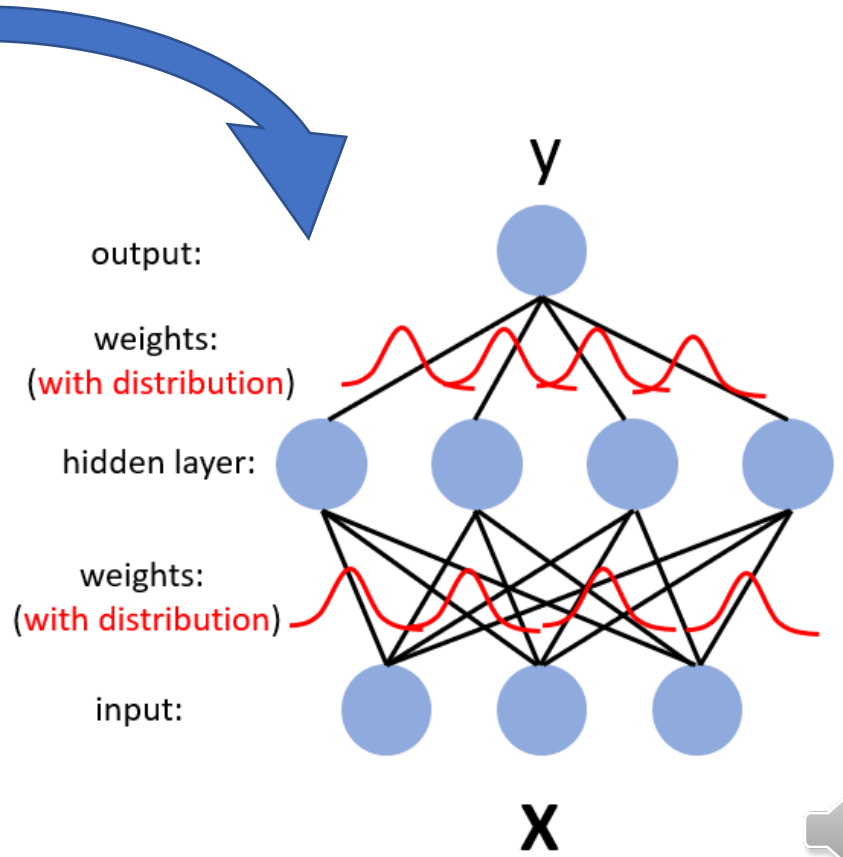


Simple&Effective Techniques: Stochastic Regularization

BatchNorm as Variational Inference



$$\widehat{x}^{(k)} = \frac{x^{(k)} - \mathrm{E}[x^{(k)}]}{\sqrt{\mathrm{Var}[x^{(k)}]}}$$



[Teye, Azizpour, Smith, "Bayesian Uncertainty Estimation for Batch Normalized Deep Networks", ICML 2018]

Simple&Effective Techniques: Stochastic Regularization

BatchNorm as Variational Inference: idea

- A few more assumptions
  - factorized prior
  - monte carlo expectation
  - weights of different layers are independent
  - batch is large enough for the central limit theorem to come in (fine with >10)
  - …

→Equivalent to variational approximation with a zero-mean Gaussian prior on BatchNorm parameters

What does this mean?

[Teye, Azizpour, Smith, "Bayesian Uncertainty Estimation for Batch Normalized Deep Networks", ICML 2018]

Simple&Effective Techniques: Stochastic Regularization

BatchNorm as Variational Inference: idea

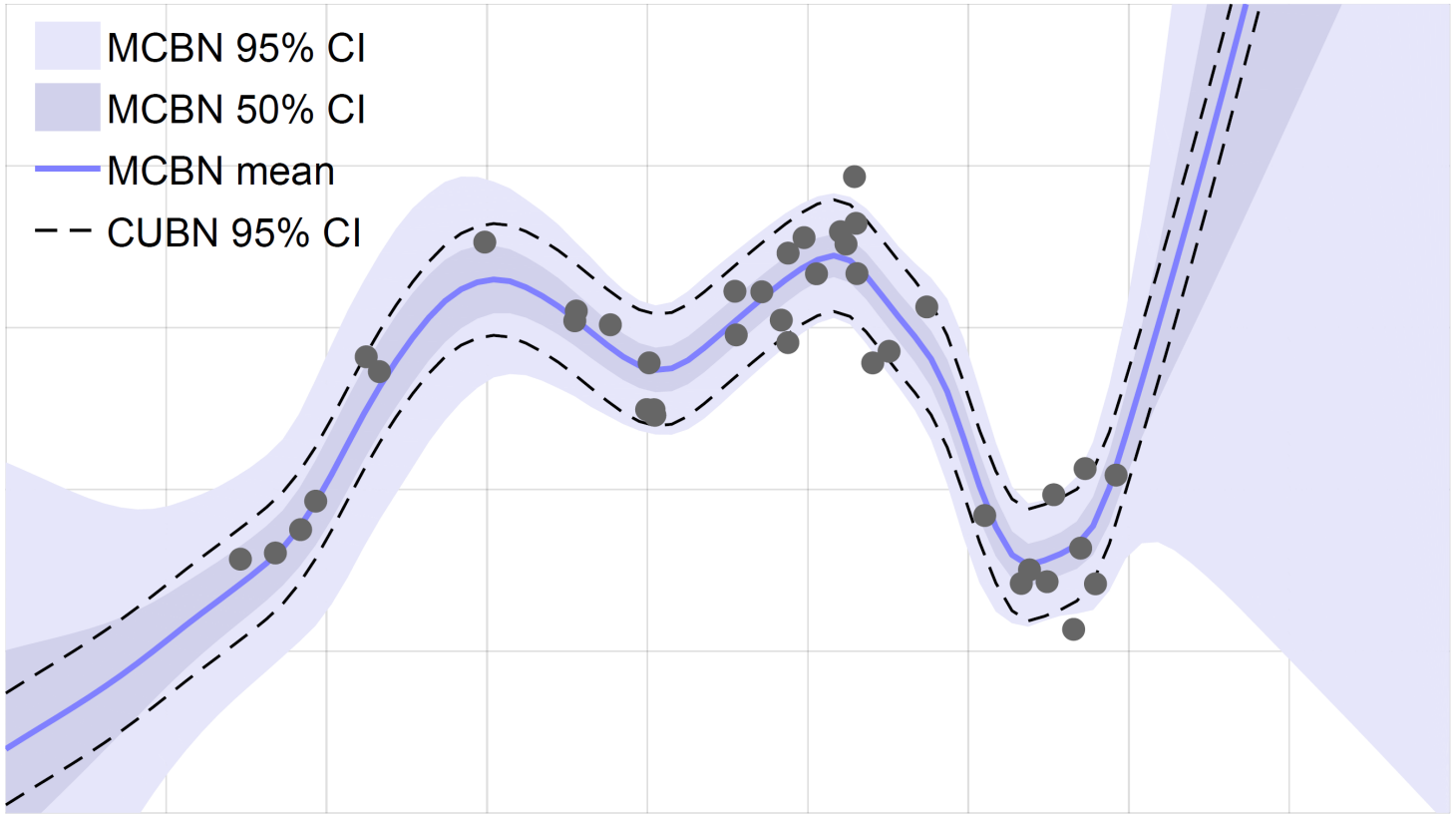
$$P(y|x,D) = \int P(y|x,D,\theta)P(\theta|D,x)d\theta = \int P(y|x,\theta)P(\theta|D)d\theta$$

- Variational inference for batchnorm networks:
  1. Sample S mini-batches from training set with i.i.d. members
  2. Calculate μ_k and σ_k for each BN unit
  3. Evaluate the network S times with $\{f(\boldsymbol{x};\boldsymbol{\theta},\boldsymbol{\mu}_i,\boldsymbol{\sigma}_i)\}_{i=1:S}$
  4. The average is the mean prediction
     - $\hat{y} = E_i[f(\boldsymbol{x};\boldsymbol{\theta},\boldsymbol{\mu}_i,\boldsymbol{\sigma}_i)]$
  5. The variance is the uncertainty
     - $\hat{\sigma}^2 = E_i[(f(\boldsymbol{x};\boldsymbol{\theta},\boldsymbol{\mu}_i,\boldsymbol{\sigma}_i) - \hat{y})^2]$

[Teye, Azizpour, Smith, "Bayesian Uncertainty Estimation for Batch Normalized Deep Networks", ICML 2018]

Simple&Effective Techniques: Stochastic Regularization

BatchNorm as Variational Inference: Results



[Teye, Azizpour, Smith, "Bayesian Uncertainty Estimation for Batch Normalized Deep Networks", ICML 2018]

Simple&Effective Techniques: Stochastic Regularization

BatchNorm as Variational Inference: Results



[Teye, Azizpour, Smith, “Bayesian Uncertainty Estimation for Batch Normalized Deep Networks”, ICML 2018]

Simple&Effective Techniques: Stochastic Regularization

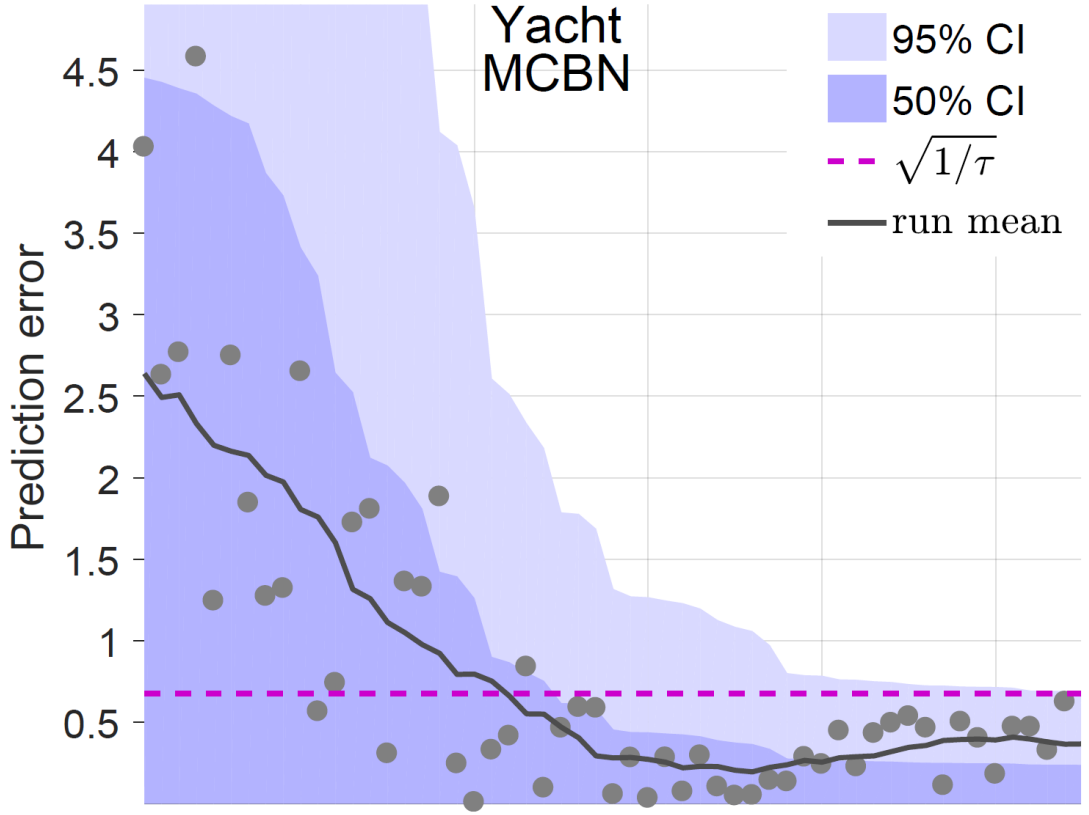
BatchNorm as Variational Inference: Results

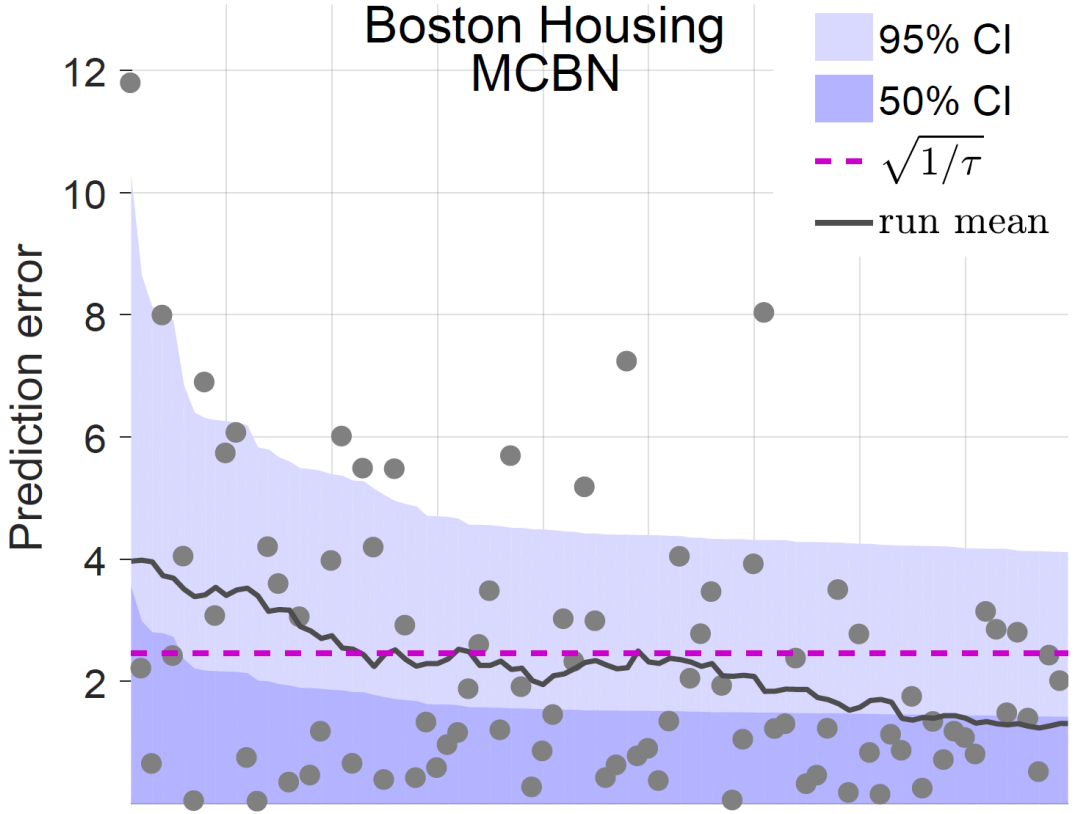
Dataset	$\overline{\text{CRPS}}$ MCBN	MCDO	MNF	$\overline{\text{PLL}}$ MCBN	MCDO	MNF
Boston	8.50 ****	3.06 ****	5.88 ****	10.49 ****	5.51 ****	1.76 ns
Concrete	3.91 ****	0.93 *	3.13 ***	-36.36 **	10.92 ****	-2.16 ns
Energy	5.75 ****	1.37 ns	1.10 ns	10.89 ****	-14.28 *	-33.88 ns
Kin8nm	2.85 ****	1.82 ****	0.53 ns	1.68 ***	-0.26 ns	0.42 ns
Power	0.24 ***	-0.44 ****	-0.89 ****	0.33 **	3.52 ****	-0.87 ****
Protein	2.66 ****	0.99 ****	0.57 ****	2.56 ****	6.23 ****	0.52 ****
Wine (Red)	0.26 **	2.00 ****	0.94 ****	0.19 *	2.91 ****	0.83 ****
Yacht	-56.39 ***	21.42 ****	24.92 ****	45.58 ****	-41.54 ns	46.19 ****

[Teye, Azizpour, Smith, "Bayesian Uncertainty Estimation for Batch Normalized Deep Networks", ICML 2018]

Simple&Effective Techniques: Stochastic Regularization

BatchNorm as Variational Inference: Results





[Teye, Azizpour, Smith, "Bayesian Uncertainty Estimation for Batch Normalized Deep Networks", ICML 2018]

SGD itself is a stochastic aspect of training!

Remember the weight noise approach?

Remember sampling as an approximation technique?



Simple&Effective Techniques: Sampling

Sampling-based Techniques

- MCMC: Generate samples from a posterior iteratively
  - Huge number of parameters in deep networks → Storage problems (maybe coresets, Huggins et al. NIPS 2016)
  - Like other MC methods, setting its hyperparameters has its own tricks of the trade on top of neural networks!
  - Does not scale to large data, requires full gradient of the likelihood on the whole dataset

- Langevin dynamics
  - Stochastic gradients of the likelihood
  - SGLD generates sample from the model's posterior for the model's random variables $\boldsymbol{\theta}_t$ by adding stochastic gradient steps to the previously generated samples

[Welling&Teh, "Bayesian Learning via Stochastic Gradient Langevin Dynamics", ICML 2011]

[Mandt et al. "Stochastic Gradient Descent as Approximate Bayesian Inference", JMLR 2017]